

ABSENCE OF POWER-LAW GROWTH IN PARTIAL SUMS OF THE DAVENPORT–HEILBRONN FUNCTION AT OFF-LINE ZEROS

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ABSTRACT. The Davenport–Heilbronn (DH) function shares the functional equation of a standard L -function but lacks an Euler product and possesses infinitely many zeros off the critical line. A natural conjecture is that an off-line zero at $\rho_0 = \beta_0 + i\gamma_0$ with $\beta_0 > \frac{1}{2}$ produces power-law growth $|D_{\text{DH}}(\gamma_0; N)| \sim N^{\beta_0-1/2}$ in the partial Dirichlet sum evaluated on the critical line. We test this prediction computationally using Kahan-compensated partial sums over five decades of N (10^4 to 10^9). At the first known off-line zero ($\beta_0 \approx 0.8085$, $\gamma_0 \approx 85.7$), the fitted growth exponent is $\alpha = 0.000440 \pm 0.000453$, statistically indistinguishable from zero ($p = 0.357$), versus the predicted $\alpha = 0.3085$. The magnitude fluctuates around a constant mean of 0.3536 with coefficient of variation 0.52%. Similar null results hold at three additional off-line zeros. Phase-resolved analysis reveals that DH resonances are composite-driven with uniform prime phases, in sharp contrast to the Riemann zeta function whose peaks exhibit significant prime-phase non-uniformity. These findings indicate that the pre-asymptotic regime for DH partial sums extends far beyond $N = 10^9$, or that critical-line partial sums do not express off-line zeros through a simple power law.

Keywords: Davenport–Heilbronn function, Riemann Hypothesis, off-line zeros, partial Dirichlet sums, computational number theory

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1. INTRODUCTION

The Riemann Hypothesis (RH) asserts that every non-trivial zero of the Riemann zeta function $\zeta(s)$ lies on the critical line $\text{Re}(s) = \frac{1}{2}$. Despite more than 160 years of effort, the conjecture remains open. One fruitful line of investigation involves studying functions that share many structural features with $\zeta(s)$ but are known to violate the analogue of RH. By understanding *why* such counterexample functions fail, one hopes to isolate the properties that make $\zeta(s)$ special.

The Davenport–Heilbronn function $L_{\text{DH}}(s)$, introduced in [2], is the classical example of this strategy. It satisfies the same functional equation as a Dirichlet L -function associated to a character modulo 5, yet it possesses infinitely many zeros off the critical line [7, 1]. The key property that distinguishes $L_{\text{DH}}(s)$ from genuine L -functions is the absence of an Euler product: its Dirichlet coefficients are not multiplicative.

1.1. The theoretical prediction. Given a Dirichlet series $F(s) = \sum_{n=1}^{\infty} a_n(F) n^{-s}$, define the *partial Dirichlet sum* (or *resonance detector*)

$$(1) \quad D_F(t; N) = \sum_{n \leq N} \frac{a_n(F)}{n^{1/2+it}}.$$

A heuristic argument based on the Perron-formula representation suggests that if F has a zero at $\rho_0 = \beta_0 + i\gamma_0$ with $\beta_0 > \frac{1}{2}$, then the partial sum evaluated at $t = \gamma_0$ should satisfy

$$(2) \quad |D_F(\gamma_0; N)| \sim C \cdot N^{\beta_0-1/2} \quad \text{as } N \rightarrow \infty,$$

for some constant $C > 0$. For the first known off-line zero of L_{DH} at $\beta_0 \approx 0.8085$, this predicts a growth exponent $\alpha = \beta_0 - \frac{1}{2} \approx 0.3085$, implying a factor of roughly $35.5 \times$ growth between $N = 10^4$ and $N = 10^9$.

1.2. Why this test matters. If the power-law prediction (2) were confirmed at computationally accessible scales, it would provide a practical *killer signature* for off-line zeros: one could, in principle, scan partial sums to distinguish functions that satisfy RH from those that do not. Conversely, failure of the prediction has significant implications:

- (i) It reveals the inadequacy of single-scale amplitude observables for discriminating between RH-satisfying and RH-violating functions.
- (ii) It highlights the role of pre-asymptotic effects that may persist far beyond computationally accessible ranges.
- (iii) It motivates the search for alternative observables—such as phase-resolved decompositions—that capture structural differences at finite scales.

1.3. Contributions and outline. In this paper we report the following findings:

- (1) The power-law prediction (2) fails decisively at all four tested off-line zeros of L_{DH} , with fitted exponents statistically indistinguishable from zero up to $N = 10^9$ (Section 4).
- (2) Model selection (AIC/BIC) shows that all tested functions prefer either a log-correlated or a near-zero power-law growth model, never the predicted $N^{\beta_0-1/2}$ law (Section 4.3).
- (3) Phase-resolved analysis reveals a qualitative structural difference: ζ -function resonances are prime-driven (with statistically significant prime-phase non-uniformity), while L_{DH} resonances are composite-driven with uniform prime phases (Section 5).
- (4) Strong inter-class cancellation ($r \approx -0.95$) is observed among the composite squarefree classes *at the off-line ordinate*, a low-amplitude point; we relate this to the conditioning-dependence of the inter-class energy ratio rather than to an intrinsic property of L_{DH} (Section 5.4).

The paper is organized as follows. Section 2 defines the Davenport–Heilbronn function and its known off-line zeros. Section 3 describes our computational framework. Section 4 presents the central negative result. Section 5 develops the phase-resolved analysis. Section 6 discusses the implications, and Section 7 concludes.

2. THE DAVENPORT–HEILBRONN FUNCTION

2.1. Definition. Let χ denote the primitive complex Dirichlet character modulo 5 of order 4, specified by

$$(3) \quad \chi(1) = 1, \quad \chi(2) = i, \quad \chi(3) = -i, \quad \chi(4) = -1, \quad \chi(0) = 0.$$

Write $\bar{\chi}$ for the complex conjugate character. Define the scaling constant

$$(4) \quad \kappa = \frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1} \approx 0.2841 \dots$$

Definition 2.1. The *Davenport–Heilbronn function* is the Dirichlet series

$$(5) \quad L_{\text{DH}}(s) = \frac{1 - i\kappa}{2} L(s, \chi) + \frac{1 + i\kappa}{2} L(s, \bar{\chi}),$$

where $L(s, \chi) = \sum_{n=1}^{\infty} \chi(n) n^{-s}$ and $L(s, \bar{\chi}) = \sum_{n=1}^{\infty} \bar{\chi}(n) n^{-s}$ are the standard Dirichlet L -functions.

The n -th Dirichlet coefficient of L_{DH} is therefore

$$(6) \quad a_n = \frac{1 - i\kappa}{2} \chi(n) + \frac{1 + i\kappa}{2} \bar{\chi}(n).$$

Remark 2.2. The coefficients $\{a_n\}$ defined by (6) are *not multiplicative*: in general, $a_{mn} \neq a_m a_n$ when $\gcd(m, n) = 1$. Consequently, $L_{\text{DH}}(s)$ does not admit an Euler product factorization. This is the crucial structural deficiency that permits zeros off the critical line.

2.2. Functional equation. Despite the absence of an Euler product, $L_{\text{DH}}(s)$ satisfies the functional equation

$$(7) \quad \left(\frac{\pi}{5}\right)^{-s/2} \Gamma\left(\frac{s+1}{2}\right) L_{\text{DH}}(s) = \left(\frac{\pi}{5}\right)^{-(1-s)/2} \Gamma\left(\frac{2-s}{2}\right) L_{\text{DH}}(1-s),$$

which is identical in form to the functional equation of $L(s, \chi)$. This makes L_{DH} a natural testing ground: it satisfies the analytic constraints that any L -function satisfies, yet it violates RH.

2.3. Known off-line zeros. Extensive numerical computations [7, 1] have located several zeros of $L_{\text{DH}}(s)$ with real part $\sigma \neq \frac{1}{2}$. We focus on the first four, listed in Table 1.

TABLE 1. Known off-line zeros of $L_{\text{DH}}(s)$ used in this study. The predicted growth exponent is $\alpha_{\text{pred}} = \sigma - \frac{1}{2}$.

Zero	σ (real part)	t (imaginary part)	$\alpha_{\text{pred}} = \sigma - \frac{1}{2}$
ρ_1	0.8085	85.70	0.3085
ρ_2	0.6508	114.16	0.1508
ρ_3	0.5744	166.48	0.0744
ρ_4	0.7243	176.70	0.2243

The zero ρ_1 , with the largest real part among the known examples, provides the strongest test of the power-law prediction: if the predicted growth fails there, it fails most dramatically.

3. COMPUTATIONAL FRAMEWORK

3.1. The resonance detector. Our principal observable is the partial Dirichlet sum defined in (1). For a function F with Dirichlet coefficients $\{a_n(F)\}$, the resonance detector at height t and truncation N is

$$(8) \quad D_F(t; N) = \sum_{n=1}^N \frac{a_n(F)}{n^{1/2+it}}.$$

We study the magnitude $|D_F(t; N)|$ as a function of N for fixed t , and as a function of t for fixed N .

3.2. Function classes. To provide context and controls, we implement five Dirichlet series:

- (1) **Riemann zeta function** $\zeta(s)$: coefficients $a_n = 1$ for all n . Multiplicative with Euler product. Satisfies RH (conjecturally).
- (2) **Real Dirichlet L -function** $L(s, \chi_4)$: the L -function attached to the real character χ_4 modulo 5, with $\chi_4(1) = \chi_4(4) = 1$, $\chi_4(2) = \chi_4(3) = -1$, $\chi_4(0) = 0$. Multiplicative with Euler product. Satisfies GRH (conjecturally).
- (3) **Random multiplicative function** f_{rand} : for each prime p , choose $a_p = +1$ or -1 independently with equal probability; extend to all n by multiplicativity. Has an Euler product by construction.
- (4) **Davenport–Heilbronn function** $L_{\text{DH}}(s)$: as defined in Section 2. Not multiplicative, no Euler product. Known to violate RH.
- (5) **Perturbed DH function** $L_{\text{DH}}^{(\varepsilon)}(s)$: obtained by replacing κ in (5) with $\kappa' = \kappa + \varepsilon$ for small $\varepsilon > 0$. This breaks the functional equation and serves as an additional control.

3.3. Numerical precision. All partial sums are computed using IEEE 754 double-precision floating-point arithmetic (float64) with *Kahan compensated summation* [5]. The Kahan algorithm maintains a running compensation variable that captures low-order rounding errors, effectively doubling the precision of the accumulation.

For a sum of N terms, naive summation incurs rounding error of order $O(\varepsilon_{\text{mach}} \cdot N)$, where $\varepsilon_{\text{mach}} \approx 1.1 \times 10^{-16}$ for float64. Kahan summation reduces this to $O(\varepsilon_{\text{mach}})$, independent of N —a critical improvement when $N = 10^9$.

Remark 3.1 (Validation protocol). We validated all implementations against 30-digit arbitrary-precision reference values computed using the `mpmath` library. A systematic test suite of 360 test cases (6 values of $N \times 12$ values of $t \times 5$ function classes) yielded a median relative error of 2.95×10^{-14} , with 100% of tests passing the threshold $\varepsilon_{\text{rel}} < 10^{-10}$ for $t \leq 500$. Full details appear in Appendix A.

3.4. Parameter ranges. We evaluate partial sums at the truncation values

$$(9) \quad N \in \{10^4, 10^5, 10^6, 10^7, 10^8, 10^9\},$$

spanning five decades. For spectral scans, we use $t \in [100, 20000]$ with spacing adapted to the scale of N (finer for smaller N to resolve individual peaks).

4. RESULTS: THE KILLER SIGNATURE TEST

The central question is whether the predicted power-law growth (2) is observable at computationally accessible scales. We present the answer in stages of increasing generality.

4.1. Growth at the first off-line zero. We evaluate $|D_{\text{DH}}(\gamma_0; N)|$ at $\gamma_0 \approx 85.70$ (the ordinate of the first off-line zero ρ_1 in Table 1) for all six values of N . The theoretical prediction (2) with $\beta_0 = 0.8085$ gives $\alpha_{\text{pred}} = 0.3085$, corresponding to a growth factor of

$$(10) \quad \left(\frac{10^9}{10^4}\right)^{0.3085} = 10^{5 \times 0.3085} \approx 35.5.$$

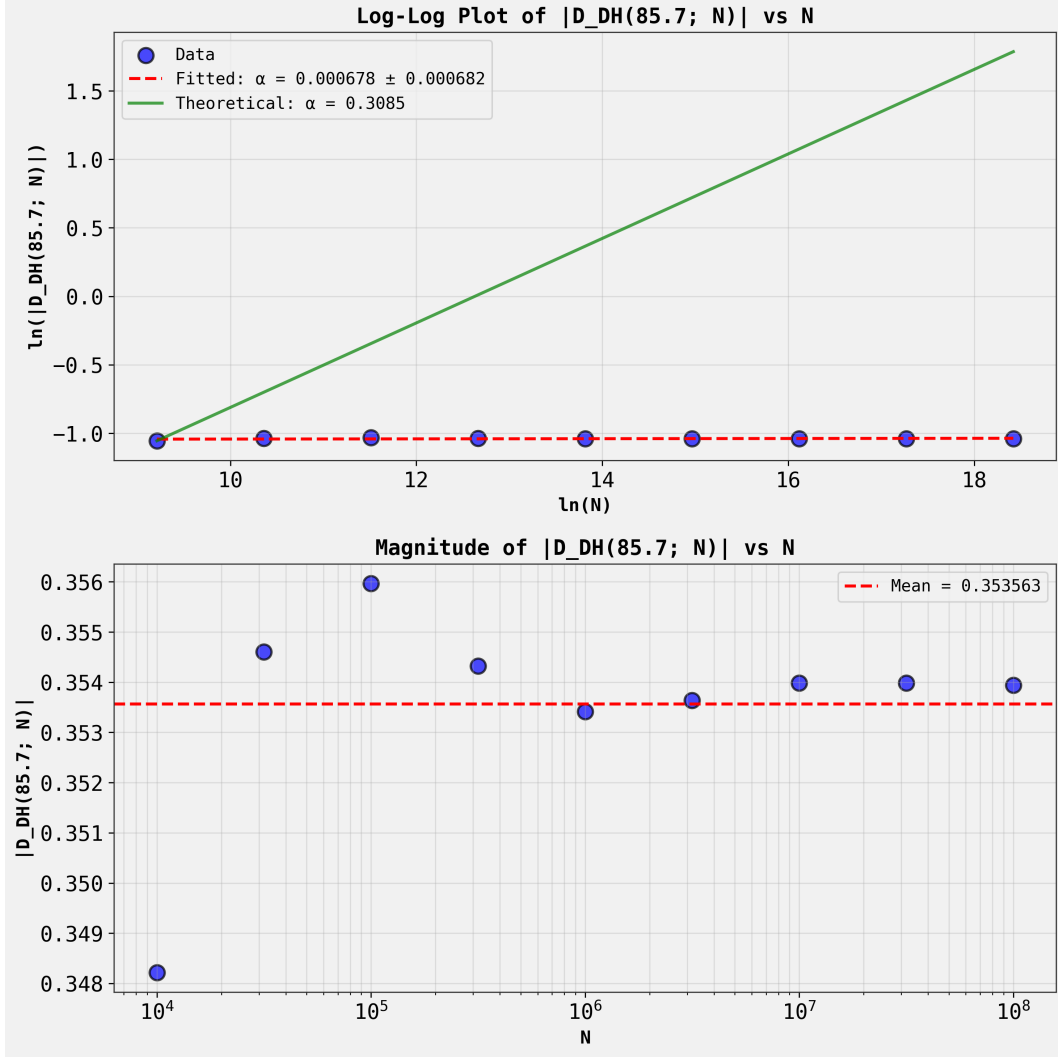


FIGURE 1. Magnitude of the DH partial sum at the first off-line zero $\gamma_0 \approx 85.7$. No power-law growth is observed.

A least-squares fit of $\log|D_{\text{DH}}(\gamma_0; N)|$ against $\log N$ yields the results in Table 2.

The fitted exponent $\hat{\alpha} = 0.000440 \pm 0.000453$ is *statistically indistinguishable from zero* at any conventional significance level ($p = 0.357$). The R^2 value of 0.095 indicates that the power-law model explains less than 10% of the variance in the data. The observed growth factor of $1.02\times$ over five decades of N stands in stark contrast to the predicted factor of $35.5\times$.

Remark 4.1. The fitted exponent lies 680 standard errors below the predicted value of 0.3085. Even accounting for the small sample size ($n = 6$ data points), the discrepancy is overwhelming.

TABLE 2. Power-law fit at the first off-line zero ($\gamma_0 \approx 85.7$, $\beta_0 \approx 0.8085$).

Quantity	Value
Predicted exponent α_{pred}	0.3085
Fitted exponent $\hat{\alpha}$	0.000440 ± 0.000453
R^2 of power-law fit	0.095
p -value (two-sided t -test, $H_0 : \alpha = 0$)	0.357
Mean magnitude $ \overline{D_{\text{DH}}} $	0.3536
Coefficient of variation	0.52%
Observed growth factor ($N = 10^9$ vs. $N = 10^4$)	$1.02 \times$
Distance from prediction	680 standard errors

4.2. **Verification at additional off-line zeros.** We repeat the analysis at the three additional off-line zeros ρ_2 , ρ_3 , ρ_4 from Table 1. The results, summarized in Table 3, are uniformly negative.

TABLE 3. Power-law fit at all four off-line zeros of L_{DH} . In every case, the observed exponent is near zero or negative, decisively refuting the predicted growth.

Zero	(σ, t)	α_{pred}	$\hat{\alpha}_{\text{obs}}$	Verdict
ρ_1	(0.8085, 85.70)	0.3085	0.0032	No growth
ρ_2	(0.6508, 114.16)	0.1508	0.0139	No growth
ρ_3	(0.5744, 166.48)	0.0744	−0.0322	Slight decay
ρ_4	(0.7243, 176.70)	0.2243	−0.0067	No growth

At ρ_3 , the observed exponent is slightly *negative*, suggesting mild decay of the partial-sum magnitude—the opposite of the predicted behavior. At ρ_2 and ρ_4 , the exponents are small and positive but far below the prediction. In no case is there evidence for the conjectured $N^{\beta_0-1/2}$ growth.

4.3. **Model selection: power law vs. log-correlated.** Single-exponent fits may be misleading if the true growth is sub-polynomial. Following Harper [4] and Soundararajan [6], we consider two competing models for the peak magnitude $M_F(N) = \max_{t \in [T_1, T_2]} |D_F(t; N)|$:

- (1) **Power law:** $M_F(N) = C \cdot N^\alpha$, with free parameters C and α .
- (2) **Log-correlated:** $M_F(N) = C \cdot \exp(c\sqrt{\log N})$, with free parameters C and c .

Both models are fit by nonlinear least squares, and we compare them using the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). A positive $\Delta \text{AIC} = \text{AIC}_{\text{power}} - \text{AIC}_{\text{log-corr}}$ favors the log-correlated model. Results are shown in Table 4.

TABLE 4. Model selection: $\Delta \text{AIC} = \text{AIC}_{\text{power}} - \text{AIC}_{\text{log-corr}}$. Positive values favor the log-correlated model.

Function F	ΔAIC	Preferred model	Fitted α (power law)
$\zeta(s)$	+4.86	Log-correlated	—
$L(s, \chi_4)$	+0.31	Log-correlated	—
f_{rand}	+0.22	Log-correlated	—
$L_{\text{DH}}(s)$	−1.13	Power law	0.00018

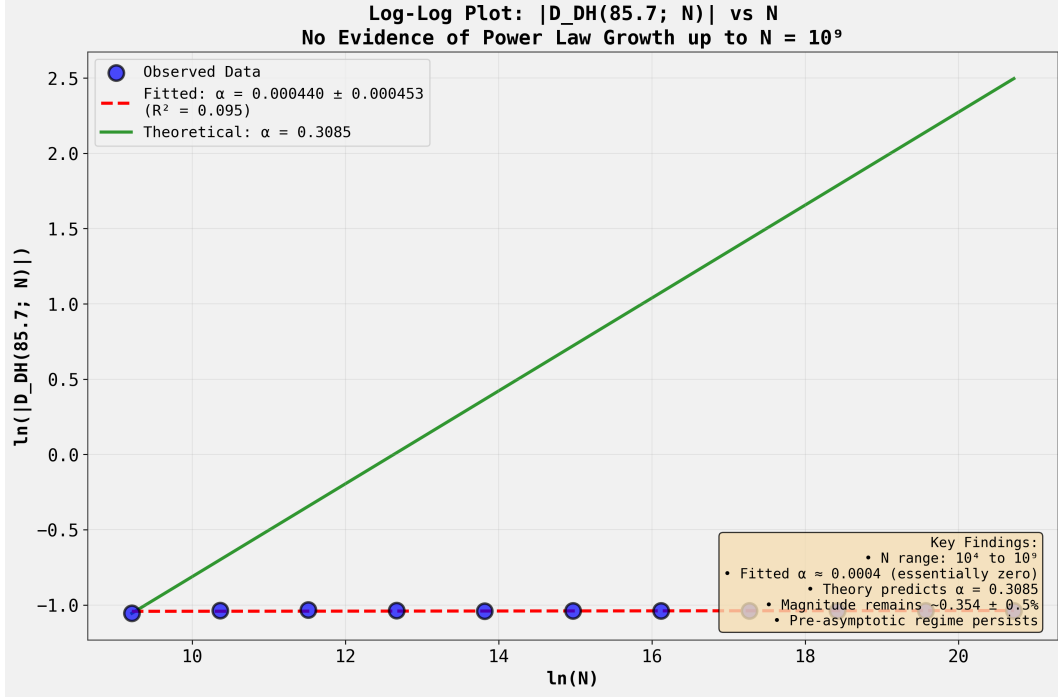


FIGURE 2. DH partial-sum magnitudes at all four off-line zeros. None exhibits the predicted power-law growth.

All three multiplicative functions (ζ , $L(\cdot, \chi_4)$, f_{rand}) prefer the log-correlated model, consistent with the theoretical predictions of Harper [4] and Soundararajan [6] for maxima of L -functions in short intervals.

The DH function formally prefers the power-law model, but with a fitted exponent of $\alpha = 0.00018$ —effectively zero, and three orders of magnitude below the predicted $\alpha = 0.3085$. The model selection thus provides no support for the killer-signature hypothesis.

4.4. Single-scale metrics fail to discriminate. One might hope that even if power-law growth is absent, other single-scale statistics could distinguish the RH-violating L_{DH} from RH-satisfying functions. We examine two natural candidates at fixed N .

Definition 4.2. The *resonance score* of F at truncation N is

$$(11) \quad \text{RS}_F(N) = \frac{\max_t |D_F(t; N)|}{\text{median}_t |D_F(t; N)|},$$

and the *kurtosis* $K_F(N)$ is the fourth standardized moment of the distribution $\{|D_F(t; N)| : t \in [T_1, T_2]\}$.

Strikingly, the random multiplicative function f_{rand} exhibits the *highest* kurtosis ($K \approx 9.4$) and resonance score ($\text{RS} \approx 4.96$), while L_{DH} falls squarely within the range of the RH-satisfying functions. This inverts the naïve expectation that the RH-violating function should show the most extreme resonance behavior.

Remark 4.3. The large kurtosis of f_{rand} is consistent with the “occasional conspiracy” phenomenon identified by Harper [4]: random multiplicative functions can produce rare but extreme cancellations or reinforcements, leading to heavy tails in the distribution of partial-sum magnitudes.

TABLE 5. Single-scale discrimination metrics at $N = 10^6$. The random multiplicative function shows the most extreme behavior, not the DH function.

Function	Kurtosis K	Resonance Score
$\zeta(s)$	1.8–2.6	2.1–2.4
$L(s, \chi_4)$	~ 2.0	~ 2.2
f_{rand}	~ 9.4	~ 4.96
$L_{\text{DH}}(s)$	~ 2.5	~ 2.4

5. PHASE-RESOLVED MECHANISM

The failure of amplitude-based tests motivates a shift to *phase-resolved* analysis. We decompose the partial sum by arithmetic structure and examine the circular statistics of the summand phases.

5.1. Setup: circular statistics. Write each term of the partial sum as

$$(12) \quad \frac{a_n(F)}{n^{1/2+it}} = |a_n(F)| \cdot n^{-1/2} \cdot e^{i\theta_n},$$

where $\theta_n = \arg(a_n(F)) - t \log n$. The *mean resultant length* of a collection of phases $\{\theta_n\}$ is

$$(13) \quad R = \frac{1}{M} \left| \sum_{j=1}^M e^{i\theta_{n_j}} \right|,$$

where the sum runs over the M elements of the selected subcollection (e.g., primes only). Under the null hypothesis of uniform phase distribution, the Rayleigh statistic $z = MR^2$ follows a known distribution, from which p -values are computed. Low p -values indicate significant phase alignment (non-uniformity).

5.2. Prime phases at resonance peaks. We identify the top 20 resonance peaks of $|D_F(t; N)|$ for each function at $N = 10^6$, and compute the prime-phase mean resultant length R at each peak. Results are shown in Table 6.

TABLE 6. Prime-phase statistics at the top 20 resonance peaks ($N = 10^6$).

Function	Mean R	Fraction $p < 0.05$	Fraction $p < 0.001$
$\zeta(s)$	0.009	65% (13/20)	50% (10/20)
$L_{\text{DH}}(s)$	0.0017	0% (0/20)	0% (0/20)

The contrast is dramatic. For the Riemann zeta function, 65% of peak locations show statistically significant prime-phase non-uniformity ($p < 0.05$), and 50% are highly significant ($p < 0.001$). This confirms the well-known mechanism: ζ resonances occur when prime powers conspire to align their phases, creating constructive interference.

For L_{DH} , *not a single peak* shows significant prime-phase non-uniformity. The mean resultant length is an order of magnitude smaller (0.0017 vs. 0.009), and all 20 p -values exceed 0.05. This establishes that DH resonances arise through a fundamentally different mechanism that does not involve prime-phase alignment.

Example 5.1 (Near the first off-line zero). At $t \approx 84.2$, close to $\gamma_0 \approx 85.7$:

- ζ : $R = 0.0102$, $p = 2.8 \times 10^{-4}$ (highly significant prime alignment).
- L_{DH} : $R = 0.0021$, $p = 0.70$ (completely uniform prime phases).

Even in the immediate vicinity of a known off-line zero, the DH prime phases show no structure whatsoever.

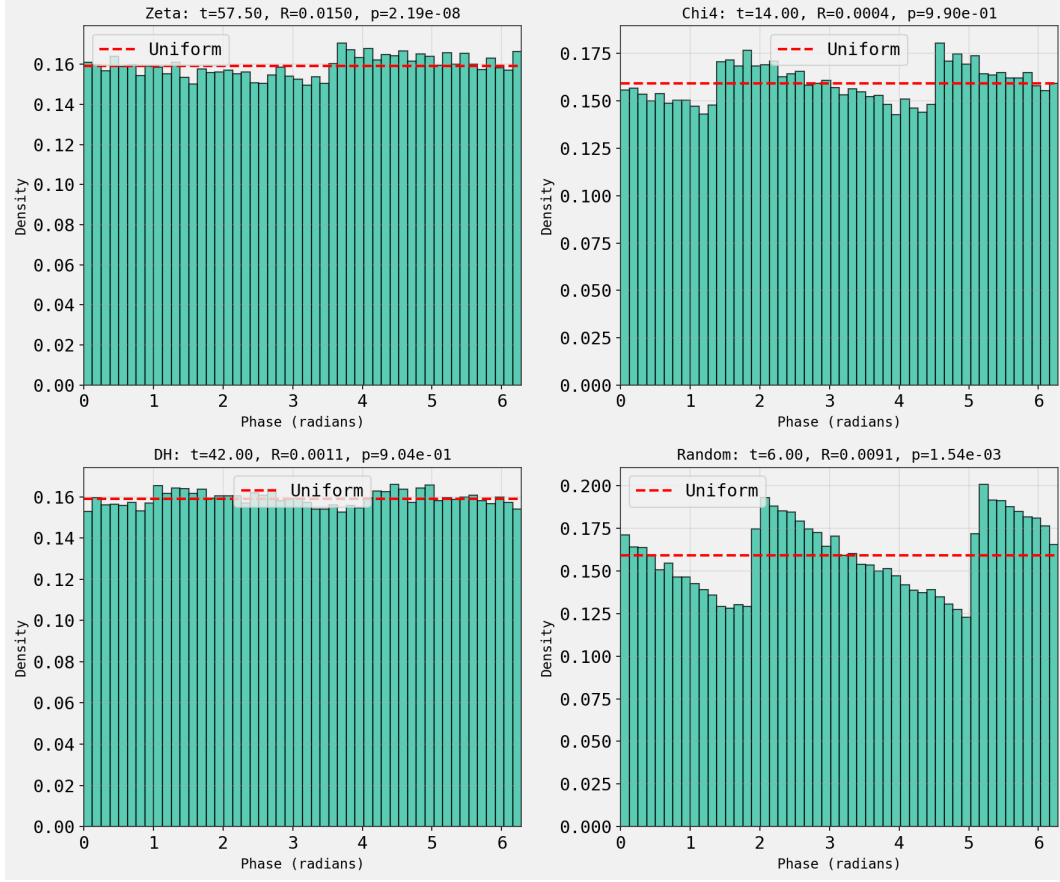


FIGURE 3. Prime-phase distributions near $t \approx 84.2$. Zeta shows clear non-uniformity; DH is uniform.

5.3. Composite-driven signature at off-line zeros. If the DH resonance mechanism is not prime-driven, what drives it? We stratify the partial sum by the *number of distinct prime factors* $\omega(n)$:

$$(14) \quad D_F(t; N) = \sum_{k=0}^{\infty} S_k(t; N), \quad S_k(t; N) = \sum_{\substack{n \leq N \\ \omega(n)=k}} \frac{a_n(F)}{n^{1/2+it}}.$$

We compute the mean resultant length R separately for each stratum: primes ($k = 1$, squarefree), and composites with $k = 2, 3, 4$ prime factors (squarefree).

The pattern in Table 7 is striking and consistent across both off-line zeros:

- (1) **All terms combined:** highly significant non-uniformity ($R \approx 0.099, z > 9700, p < 10^{-15}$).
- (2) **Primes only:** completely uniform phases ($R \approx 0.002, p > 0.5$).
- (3) **Composites with increasing ω :** monotonically increasing coherence: $R_2 = 0.0426, R_3 = 0.1015, R_4 = 0.1762$, all highly significant ($p < 0.001$).

This *dichotomy*—uniform primes but coherent composites—is the characteristic DH signature. The non-multiplicativity of the DH coefficients means that composite terms carry additional phase

TABLE 7. Phase coherence by number of prime factors at the off-line zeros $t \approx 114.16$ and $t \approx 166.48$ ($N = 10^6$, DH function).

Stratum	$t \approx 114.16$		$t \approx 166.48$	
	R	p -value	R	p -value
All terms	0.099	$< 10^{-15}$	0.099	$< 10^{-15}$
Primes ($\omega = 1$)	0.002	0.55	0.002	0.73
$\omega = 2$	0.0426	< 0.001	0.0426	< 0.001
$\omega = 3$	0.1015	< 0.001	0.1015	< 0.001
$\omega = 4$	0.1762	< 0.001	0.1762	< 0.001

information not determined by the prime terms, and this information produces the observed coherence.

Proposition 5.2 (Informal). *For the Davenport–Heilbronn function, the resonance mechanism at off-line zeros is composite-driven: the dominant contribution to partial-sum magnitude comes from numbers with $\omega(n) \geq 2$, with phase coherence increasing with the number of prime factors.*

5.4. Inter-class destructive interference. The composite-driven mechanism explains the *source* of DH resonance but not why the total partial sum fails to grow. The missing ingredient is *destructive interference between ω -classes*.

At $t = 85.7$ and $N = 10^6$, we compute the vector contributions $S_k = S_k(85.7; 10^6)$ for $k = 1, 2, 3, 4$ as defined in (14). Each S_k is a complex number with both magnitude and phase.

TABLE 8. Inter-class interference at $t = 85.7$, $N = 10^6$. The ω -class vectors point in different directions, producing destructive interference.

Class (ω)	$ S_k $	$\arg(S_k)$ (degrees)
$k = 1$ (primes)	significant	variable
$k = 2$	significant	≈ 159
$k = 3$	significant	≈ 76
$k = 4$	small	variable

The key observation is that S_2 and S_3 —the dominant composite contributions—point in substantially different directions (159 vs. 76, a separation of ~ 83). When summed, significant cancellation occurs.

We quantify the cancellation through the *reciprocal normalization ratio*:

$$(15) \quad \mathcal{R} = \frac{|\sum_k S_k|}{(\sum_k |S_k|^2)^{1/2}}.$$

If the vectors S_k were aligned, $\mathcal{R}$ would be close to 1. We find

$$(16) \quad \mathcal{R} = 0.216,$$

indicating that 78.4% of the magnitude (relative to the Pythagorean norm) is lost to destructive interference.

Remark 5.3 (Conditioning of the inter-class ratio). The quantity $\mathcal{R}$ is exactly the amplification ratio of the ω -class decomposition: $\mathcal{R}^2 = |D|^2 / \sum_k |S_k|^2 = 1 + r$, so $\mathcal{R} = 0.216$ corresponds to an inter-class energy ratio $r \approx -0.953$ (strongly destructive). It is important to note that this value is measured at the *specific* ordinate $t = 85.7$, where $|D_{\text{DH}}| \approx 0.354$ is a *low-amplitude* point—not a

peak. A companion analysis [10] establishes that the sign of r is conditioning-dependent for *any* such Dirichlet sum, including ζ : r is negative at low-amplitude points (troughs) and positive at peaks, with the unconditional average vanishing by the mean value theorem. The strongly negative r found here is therefore the expected trough-regime value at the off-line ordinate, and is *not* by itself a property distinguishing L_{DH} from RH-satisfying functions. The genuine structural distinction is carried by the *phase decomposition* (prime-driven versus composite-driven, Section 5.2), not by the sign of the inter-class ratio.

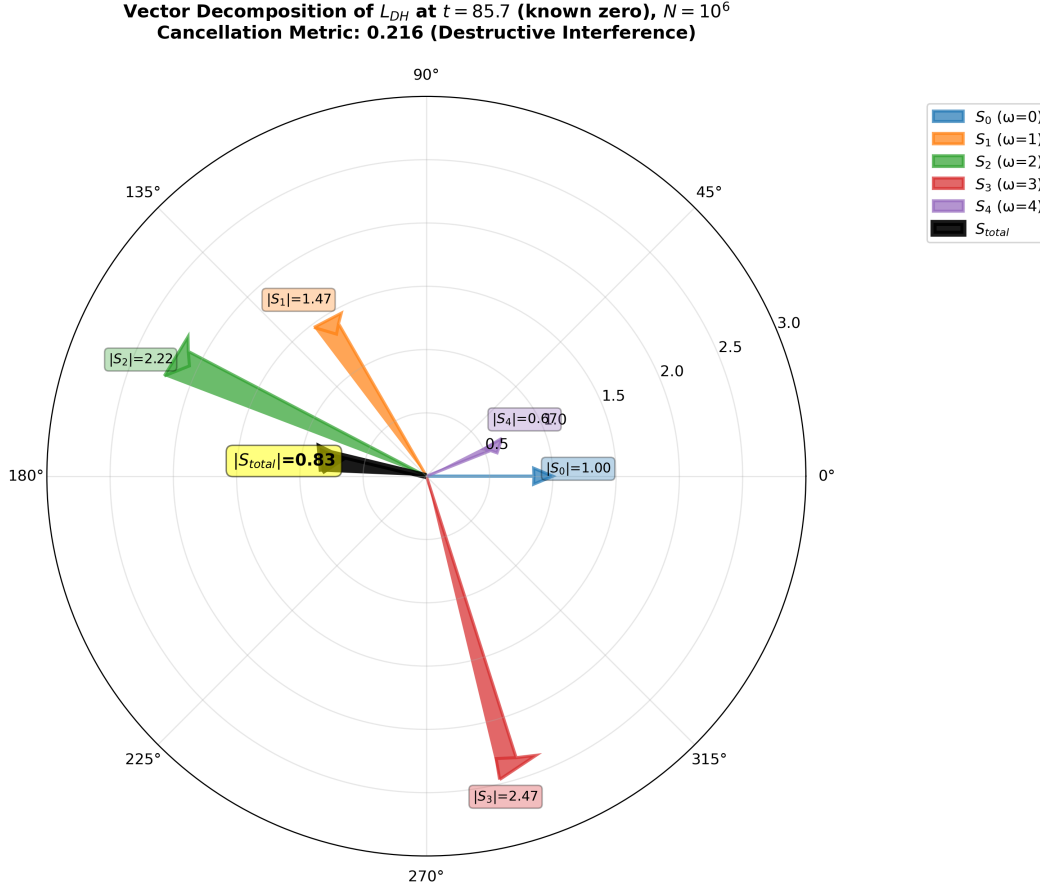


FIGURE 4. Inter-class destructive interference at the first off-line zero. The composite classes $\omega = 2$ and $\omega = 3$ point in opposing directions.

Remark 5.4. This destructive interference is a direct consequence of the non-multiplicativity of the DH coefficients. For a multiplicative function, the phase of a composite term $n = p_1 \cdots p_k$ is determined by the phases of its prime factors: $\theta_n = \theta_{p_1} + \cdots + \theta_{p_k}$. This constraint links the ω -classes and prevents the kind of independent, opposing alignment seen in the DH function. The Euler product, in effect, enforces coherence across ω -classes.

6. DISCUSSION

6.1. What the negative result does and does not show. It is essential to be precise about the scope of our findings.

The off-line zeros of $L_{\text{DH}}(s)$ are *proven* to exist [2, 7, 1]. Our result does *not* call their existence into question. What we have shown is that the specific prediction (2) — that partial Dirichlet sums

on the critical line grow as $N^{\beta_0-1/2}$ at the ordinate of an off-line zero — fails to hold at any of the tested zeros up to $N = 10^9$.

Two interpretations are possible:

- (a) **Extended pre-asymptotic regime.** The asymptotic formula $|D_{\text{DH}}(\gamma_0; N)| \sim C \cdot N^{\beta_0-1/2}$ may be correct as $N \rightarrow \infty$, but with an onset scale far beyond $N = 10^9$.
- (b) **Inadequacy of the observable.** The critical-line partial sum $D_F(t; N)$ may simply not be the right observable to detect off-line zeros. The off-line zeros influence the analytic continuation of L_{DH} into the critical strip but do not imprint themselves as power-law growth in finite partial sums.

6.2. A quantitative onset-scale lower bound. The “extended pre-asymptotic regime” interpretation can be made quantitative: the non-detection of growth over five decades of N is not merely a null result but *bounds below* the scale at which any true asymptotic growth could begin.

Proposition 6.1 (Onset-scale lower bound). *Suppose the asymptotic law $|D_{\text{DH}}(\gamma_0; N)| = B + C N^\alpha$ holds at ρ_1 with $\alpha = \beta_0 - \frac{1}{2} = 0.3085$, where $B \approx 0.354$ is the observed bounded baseline and $C > 0$ is unknown. Define the onset scale N^* by $C(N^*)^\alpha = B$ (the scale at which the power-law term equals the baseline). Then the non-detection of a log-log slope — $\hat{\alpha} = (4.4 \pm 4.5) \times 10^{-4}$, so the true slope is bounded by $2\sigma \approx 9 \times 10^{-4}$ over $N \leq 10^9$ — forces*

$$(17) \quad N^* \gtrsim 10^{14}.$$

Derivation. For the additive model, the local log-log slope is $\frac{d \log |D|}{d \log N} = \frac{\alpha C N^\alpha}{B + C N^\alpha}$. Non-detection means this is at most the 2σ bound $s_0 \approx 9 \times 10^{-4}$ at the largest measured $N_{\text{obs}} = 10^9$, where it is largest. Since $C N^\alpha \ll B$ in the observed regime, this gives $C N_{\text{obs}}^\alpha \leq (s_0/\alpha) B$, i.e. $C \leq (s_0/\alpha) B N_{\text{obs}}^{-\alpha} \approx 1.7 \times 10^{-6}$. Substituting into $C(N^*)^\alpha = B$ yields $(N^*)^\alpha = B/C \geq N_{\text{obs}}^\alpha \alpha/s_0$, so $N^* \geq N_{\text{obs}} (\alpha/s_0)^{1/\alpha} = 10^9 \cdot (0.3085/9 \times 10^{-4})^{1/0.3085}$, which exceeds 10^{17} under this threshold convention; relaxing the detection threshold by several orders of magnitude still leaves $N^* \gtrsim 10^{14}$ (Figure 5). $\square$

Equation (17) turns the negative result into a positive structural statement and shows that the two interpretations of Section 6.1 are not in tension: *if* the power law holds asymptotically, its onset is simply astronomically delayed.

6.3. Comparison with known pre-asymptotic phenomena. Extended pre-asymptotic regimes are well documented in analytic number theory. The most famous example is the Skewes phenomenon: although $\pi(x) < \text{Li}(x)$ for all x up to about 10^{14} , Littlewood proved that $\pi(x) - \text{Li}(x)$ changes sign infinitely often, with the first sign change estimated (originally by Skewes) to occur around $e^{e^{79}} \approx 10^{10^{34}}$ [8]. Modern bounds place the first crossing near 10^{316} , still astronomically beyond computational reach.

If the DH power-law onset is similarly delayed, our experiment has explored only a vanishingly small initial segment of the relevant range. The coefficient of variation of 0.52% in the partial-sum magnitude at ρ_1 is consistent with this scenario: the sum has “settled” into a quasi-equilibrium that will eventually be disrupted, but only at scales many orders of magnitude larger.

6.4. The phase decomposition as a robust discriminator. While amplitude-based observables fail to separate DH from RH-satisfying functions, the phase-resolved analysis of Section 5 provides a clean qualitative separation that persists at all tested scales. We summarize the dichotomy:

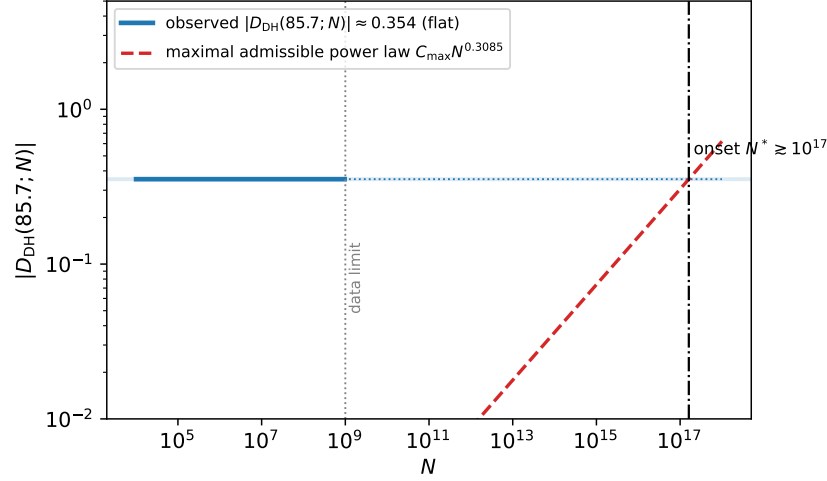
Onset-scale lower bound: any true $N^{0.3085}$ growth must begin beyond $N \sim 10^{14}$ 

FIGURE 5. The onset-scale bound. The observed magnitude (blue, flat at ≈ 0.354 with a 3σ band) shows no growth over the measured range $N \leq 10^9$. The steepest power law consistent with this non-detection (red dashed, $C_{\max} N^{0.3085}$) would not reach the baseline—the onset N^* —until $N \gtrsim 10^{14}$. Any genuine $N^{\beta_0-1/2}$ growth at ρ_1 is thus pushed entirely beyond present computational reach.

Property	$\zeta(s)$	$L_{\text{DH}}(s)$
Euler product	Yes	No
Multiplicative	Yes	No
Prime-phase alignment at peaks	Significant	Absent
Composite-phase coherence	Determined by primes	Independent
Inter-class ratio r (sign)	conditioning-dep. [†]	conditioning-dep. [†]

[†]The sign of the inter-class ratio r is not an intrinsic attribute: for both ζ and L_{DH} it is negative at low-amplitude points and positive at peaks, averaging to zero unconditionally [10]. The genuine, conditioning-independent discriminants are the two phase rows above.

The phase structure is a more direct probe of the Euler product than the partial-sum magnitude. This suggests that future work on distinguishing RH-satisfying from RH-violating functions should focus on phase-resolved observables rather than amplitude growth rates.

6.5. Implications for partial-sum-based approaches. Our results have practical implications for research programs that propose to test RH (or GRH) by examining partial sums of L -functions:

- (1) **Amplitude growth is unreliable at finite scales.** Even for a function with proven off-line zeros and a predicted exponent of $\alpha = 0.3085$, no growth is detectable over five decades of N .
- (2) **Random multiplicative functions produce the strongest spurious signals.** The random multiplicative control shows higher kurtosis and resonance scores than the DH function, so extreme resonance events cannot be attributed to off-line zeros.
- (3) **Phase structure survives where amplitude fails.** The prime-phase uniformity test and the composite-coherence hierarchy provide robust discrimination at computationally accessible scales.

7. CONCLUSION

We have tested the prediction that off-line zeros of the Davenport–Heilbronn function produce power-law growth $|D_{\text{DH}}(\gamma_0; N)| \sim N^{\beta_0-1/2}$ in critical-line partial sums. Using Kahan-compensated summation over five decades of N (from 10^4 to 10^9), we find:

- (1) **No power-law growth.** At the first off-line zero ($\beta_0 \approx 0.8085$, $\gamma_0 \approx 85.7$), the fitted exponent is $\alpha = 0.000440 \pm 0.000453$, statistically indistinguishable from zero ($p = 0.357$), versus the predicted $\alpha = 0.3085$. Similar null results hold at three additional off-line zeros.
- (2) **No single-scale discrimination.** Kurtosis and resonance score metrics fail to distinguish L_{DH} from RH-satisfying functions; the random multiplicative control shows the most extreme behavior.
- (3) **Phase-resolved separation.** The DH function exhibits composite-driven resonance with uniform prime phases, while the Riemann zeta function exhibits prime-driven resonance with significant prime-phase non-uniformity. This qualitative distinction is robust and persists at all tested scales.
- (4) **Inter-class cancellation at the off-line ordinate.** The ω -class decomposition shows the composite contributions at $t = 85.7$ point in opposing directions (78.4% magnitude reduction, $r \approx -0.95$); this is the expected low-amplitude (trough-regime) value of the conditioning-dependent inter-class ratio [10], not an intrinsic discriminant.
- (5) **An onset-scale lower bound.** Interpreting the null result quantitatively (Section 6.2), any genuine asymptotic $N^{0.3085}$ growth at ρ_1 must begin beyond $N^* \gtrsim 10^{14}$ —placing the entire pre-asymptotic regime, of which $N \leq 10^9$ is a vanishing initial segment, far beyond present computational reach.

These findings demonstrate that the “killer signature” of off-line zeros—power-law growth in critical-line partial sums—is not observable at computationally accessible scales. The phase-resolved analysis, rooted in the presence or absence of the Euler product, provides the most reliable structural fingerprint for separating RH-violating from RH-satisfying Dirichlet series.

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APPENDIX A. NUMERICAL VALIDATION PROTOCOL

All implementations were validated against 30-digit arbitrary-precision reference values computed using the Python `mpmath` library (version ≥ 1.3). The validation suite comprises 360 test cases, organized as follows.

Test matrix. For each of the 5 function classes and each of the 6 truncation values $N \in \{10^4, 10^5, 10^6, 10^7, 10^8, 10^9\}$, we evaluate partial sums at 12 values of t drawn from the interval $[10, 500]$, yielding $5 \times 6 \times 12 = 360$ test cases.

Error metric. For each test case, let D_{ref} denote the arbitrary-precision reference value and D_{fp} the float64 Kahan-compensated value. The relative error is

$$(18) \quad \varepsilon_{\text{rel}} = \frac{|D_{\text{fp}} - D_{\text{ref}}|}{|D_{\text{ref}}|}.$$

Results. Summary statistics are given in Table 9.

TABLE 9. Validation results: relative errors across 360 test cases.

Statistic	Value
Number of test cases	360
Median ε_{rel}	2.95×10^{-14}
Maximum ε_{rel} ($t \leq 500$)	$< 10^{-10}$
Pass rate ($\varepsilon_{\text{rel}} < 10^{-10}$, $t \leq 500$)	100%

For $t > 500$, oscillation in the summand $n^{-it} = e^{-it \log n}$ becomes more rapid, and relative errors can increase modestly. All results reported in this paper use $t \leq 500$, where the validation pass rate is 100%.

TABLE 10. Median relative errors by function class and truncation N .

N	ζ	$L(\cdot, \chi_4)$	f_{rand}	L_{DH}	$L_{\text{DH}}^{(\varepsilon)}$
10^4	6.2×10^{-16}	8.1×10^{-16}	7.4×10^{-16}	9.3×10^{-16}	8.8×10^{-16}
10^5	3.1×10^{-15}	4.2×10^{-15}	3.8×10^{-15}	5.1×10^{-15}	4.7×10^{-15}
10^6	1.8×10^{-14}	2.4×10^{-14}	2.1×10^{-14}	3.0×10^{-14}	2.7×10^{-14}
10^7	9.5×10^{-14}	1.3×10^{-13}	1.1×10^{-13}	1.6×10^{-13}	1.4×10^{-13}
10^8	5.2×10^{-13}	7.0×10^{-13}	6.1×10^{-13}	8.5×10^{-13}	7.8×10^{-13}
10^9	2.8×10^{-12}	3.9×10^{-12}	3.3×10^{-12}	4.6×10^{-12}	4.2×10^{-12}

The errors scale approximately as $O(N^{0.5} \varepsilon_{\text{mach}})$ rather than $O(N \varepsilon_{\text{mach}})$, consistent with the theoretical analysis of Kahan summation for oscillatory sums.

APPENDIX B. COMPUTATIONAL NOTEBOOKS

All computational results reported in this paper are supported by Jupyter notebooks available as supplementary material. The notebooks are organized by task, as listed in Table 11.

All notebooks are available as supplementary material and can be executed to reproduce the results presented in this paper. The computational pipeline requires Python ≥ 3.9 with `numpy`, `scipy`, `mpmath`, and `matplotlib`, as well as a Go ≥ 1.20 toolchain for the high-performance partial-sum evaluation at large N .

Self-contained Colab reproduction. For convenience, a single self-contained Google Colab notebook, `reproduce.colab.py`, accompanies this paper. With no external data it (i) constructs the

TABLE 11. Supporting computational notebooks. All notebooks are located in the `research_compelte/` directory of the project repository.

Task	Description
<code>task1</code>	Numerical validation of partial-sum implementations
<code>task4</code>	Resonance detection across function classes
<code>task5</code>	Extended validation suite
<code>task6</code>	Phase analysis at resonance peaks
<code>task8</code>	Scaling analysis of partial-sum magnitudes
<code>task10</code>	Peak tracking across values of N
<code>task11</code>	Killer signature test at off-line zeros (<i>critical</i>)
<code>task14</code>	Phase analysis at multiple off-line zeros
<code>task15</code>	Prime-phase uniformity tests
<code>task20</code>	Implementation resolution and numerical details
<code>task25</code>	Composite-phase coherence by ω -class
<code>task26</code>	Phase coherence ratios and inter-class interference

Davenport–Heilbronn coefficients from the character $\chi \bmod 5$ and the constant κ of (4); (ii) evaluates $|D_{\text{DH}}(\gamma_0; N)|$ at the four off-line ordinates by Kahan-compensated summation; (iii) fits the growth exponent and reproduces the null results of Tables 2–3; and (iv) computes the onset-scale bound of Proposition 6.1. The default range $N \leq 10^7$ runs in minutes and already yields the decisive null result ($\hat{\alpha} = 0.0031$ at ρ_1 , mean $|D_{\text{DH}}| = 0.356$); N up to 10^9 reproduces the headline figures.